

Justify your answers and show your work. Unless stated otherwise, all questions worth 5pts.

1. [2 pts each] Answer each question.

a. Write the infinite series, $2 + 4 + 6 + 8 \cdots$, using sigma notation (i.e. using Σ).

b. Evaluate the series $\sum_{k=2}^n (2k + 1)$ for $n = 4$.

c. If $0 < a_n < b_n$ and $\{a_n\}$ diverges then what can be said about $\{b_n\}$?

d. If $\lim_{n \rightarrow \infty} \{a_n\} = 0$ then what does the Divergence Test say about $\sum_{k=0}^{\infty} a_k$?

e. When approximating $\sqrt[3]{27.5}$ with a Taylor polynomial, what function and center should be used.

Evaluate the following sequences.

2. $\lim_{n \rightarrow \infty} \left\{ \frac{n^3 - 2n^2}{5n^3 + 2n} \right\}$

3. $\lim_{n \rightarrow \infty} \left\{ \frac{0.3^n}{n^{10}} \right\}$

Evaluate the geometric series or state that it diverges.

4. $\sum_{k=0}^{\infty} 5 \frac{4^k}{3^k}$

5. $\sum_{k=2}^{\infty} 3 \left(\frac{1}{2} \right)^k$

Use the Divergence Test to determine whether the following series diverge, or state that the test is inconclusive.

6.
$$\sum_{k=1}^{\infty} \frac{k^2 - 1}{3k^2 + 10}$$

Using the Squeeze Theorem, find the limit of the following sequence or determine that the limit does not exist.

7.
$$\lim_{n \rightarrow \infty} \left\{ \frac{(-1)^n}{n} \right\}.$$

Use the Integral Test to determine whether the series converges, diverges, or state why the test cannot be used. Check that the conditions of the test are satisfied.

8. [10 pts]
$$\sum_{k=0}^{\infty} \frac{1}{x^{2/3}}.$$

Use the function $f(x) = \sqrt{x}$ to answer the following.

9. Find the Taylor polynomial of order $n = 3$ for $f(x)$ centered at 1.

10. Use the Taylor polynomial from above to approximate $\sqrt{1.05}$.

11. Compute the absolute error in the approximation assuming the exact value is given by a calculator.